

$$0) \frac{2n+1}{n-3} \quad \frac{n}{m}$$

vertical
 $x-3=0$

$$x=3$$

Horizontal
 $n=m \quad n < m$

slant
 $n > m$

$$\lim_{x \rightarrow a} f(x) = a$$

$$\lim_{x \rightarrow \infty} f(x) = \pm \infty$$

$$\lim_{x \rightarrow 3}$$

$$\begin{aligned} & \frac{2n+1}{n-3} \\ &= \frac{n(2+\frac{1}{n})}{n(1-\frac{3}{n})} \\ &= \frac{2}{1} \end{aligned}$$

Asymptotes:-

1. find vertical and horizontal asymptote :-

$$- y = \frac{1}{x}$$

$$- y = \frac{3x^2+5}{2x-1}$$

$$- y = 1 - \frac{2}{x}$$

$$- y = \frac{2x^2+3x+1}{x^2-4}$$

and check for
 slant Asymptote
 Ab.

$$- y = \frac{x^2+1}{x^2-2x}$$

$$- y = \frac{2x^2-3x+4}{x+2}$$

$$- y = \frac{x^3}{x^2-1}$$

Q

0) $y = \frac{1}{x}$

$x = 0$ yes it exists $\lim_{x \rightarrow 0} f(x) = \pm \infty$	$x < \infty$ yes it exists $\lim_{x \rightarrow \infty} f(x) = a$ $= \frac{1}{\infty} = 0$
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0 $1 - \frac{2}{x}$

$= \frac{x-2}{x}$

$$y = \frac{2x^2 - 3x + 4}{x+2}$$

$x+2 = 0$ $x = -2$ $\lim_{x \rightarrow -2} f(x) = \pm \infty$ it exists	$x < \infty$ $\lim_{x \rightarrow \infty} f(x) = a$
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$$\begin{array}{r}
 x+2 \overline{) 2x^2 - 3x + 4} \\
 \underline{2x^2 + 4x} \\
 0 - 7x + 4 \\
 \underline{- 7x - 14} \\
 0 + 18
 \end{array}$$

$$\begin{aligned}
 -3x &= (-14x) \\
 -3x - 4x & \\
 &= -7x \\
 4 &= (-14) \\
 4 + 14 & \\
 &= 18
 \end{aligned}$$

$= \frac{2x-7}{x+2}$

$= 2$

Horizontal axis

$$y = \frac{x^3}{x^2 - 1}$$

for vertical

$$x^2 - 1 = 0$$

$$x^2 = 1$$

$$x = \pm 1$$

it exist yes

$$x^2 - 1 \overline{) \begin{array}{r} x^3 \\ - x^2 - x \\ \hline 0 \end{array} }$$

= x slat asymptotes

$$0 - (-x) \\ + x$$

$$y = \sqrt{\frac{3x^2 + 5}{2x - 1}}$$

for vertical

$$2x - 1 = 0$$

$$2x = 1$$

$$x = \frac{1}{2}$$

vertical exists at $x = \frac{1}{2}$

$$2x - 1 \overline{) \begin{array}{r} \frac{1}{2}x + \frac{5}{3} \\ 3x^2 + 0 \\ - 3x^2 - \frac{3}{2}x \\ \hline 0 \end{array} } + 5$$

$$+ \frac{3}{2}x$$

$$+ \frac{1}{2}x - \frac{5}{3}$$

$$\text{slat } y = \frac{3}{2}x + \frac{4}{3}$$

$$2 \times x = 3 \\ x = \frac{3}{2}$$

$$\frac{3}{2} \times x = 2$$

$$3x = 4 \\ x = \frac{4}{3}$$

